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Enabling superior impact toughness of low-density δ-ferrite steel by dispersing ultra-fine spheroidized carbides in the ferrite/carbide composite lamellae

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ABSTRACT

The mechanism of the synergistic effect of microstructural evolution on fracture toughness in low-density δ-ferrite steel under varying rolling temperatures was systematically investigated in this study. The findings indicated that a lamellar microstructure can be formed through high-temperature rolling (850–950 °C), consisting of δ-ferrite, honeycomb martensite lamellae, and equiaxed ferrite grains which are formed by DIFT (deformation-induced ferrite phase transformation) within the honeycomb structure. Together with the concentration of interfacial stress, the equiaxed DIF (deformation induced ferrite) grains served as pre-existing defects, resulting in an impact energy of less than 7 J at room temperature (elongation less than 9 %). Meanwhile, the monotonically decreasing θ-curve indicated that deformation coordination was hampered by phase hardness mismatch (martensite/ferrite hardness difference >3.3 GPa). Conversely, a lamellar microstructure with alternating δ-ferrite lamellae and composite lamellae of ultra-fine spherical carbides (diameter <250 nm)/ferrite was formed by low-temperature rolling (650–750 °C). Under the influence of the Orowan mechanism, cracks hardly propagated through the lamellae containing ultra-fine spherical carbides, which forced cracks to deflect horizontally and extended the propagation path. At the same time, the presence of these carbides increased the curvature radius of the crack tip to reduce stress concentration, and the microvoids induced by carbides also promoted horizontal crack propagation along lamellar boundaries. These multiple mechanisms collectively enhanced material toughness, resulting in an impact absorption energy of up to 207.5 J (elongation 19.8 %). These results suggested a new pathway for the strength-toughness regulation of lightweight steel.

1. Introduction

Low-density δ-ferrite steel has served as a critical material in automotive, aerospace, and related fields because of its excellent specific strength and lightweight potential [1,2]. The steel density reduces by the addition of Al (aluminum) (approximately 1.3 % weight reduction per 1 wt% Al), which holds strategic significance for achieving weight reduction and emission reduction in transportation vehicles [3]. In conventional steels, δ-ferrite typically exhibits as fine, dispersed particles or short rods uniformly distributed in the matrix [4,5]; however, high Al content (>4 wt%) expands the δ-ferrite phase region, leading to the stable presence of coarse δ-ferrite (grain size >50 μm) at room temperature, which severely deteriorates material toughness [6,7]. This phenomenon is due to the fact that δ-ferrite, as a BCC (body-centered cubic) phase, has a limited number of slip systems and lacks phase transformation refinement mechanisms, making grain refinement and microstructure optimization difficult by traditional heat treatment [8–10]. Studies have shown that when the volume fraction of δ-ferrite exceeds 20 vol%, the impact toughness decreases by more than 60 %, which becomes a bottleneck restricting the engineering application of low-density steel [11,12].

To overcome the limitation of low toughness in low-density δ-ferrite steel, thermomechanical processing (e.g., rolling) is adopted in δ-ferrite grain refinement and microstructure configuration design. Through hot rolling, DRX (dynamic recrystallization) occurs to refine grains, with simultaneous phase interface regulation enabling the construction of a

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lamellar microstructure to balance strength, plasticity, and toughness. In recent years, lamellar structures have shown exceptional performance. Three typical microstructures are categorized to illustrate this design concept. The first category refers to the ferrite/martensite dual-phase lamellar structure. Zhang et al. [13] achieved an interphase constraint effect to inhibit localized deformation via hot rolling, obtaining excellent mechanical properties with a yield strength (YS) of 700 MPa and a TE (total elongation) of 25 %. Similarly, Cao et al. [14] prepared a ferrite/martensite dual-phase lamellar structure via hot rolling, and the structure has an impact absorption energy of 300–450 J. The second category is characterized by multi-phase lamellar structure (e.g., martensite matrix + ultra-fine retained austenite + oriented δ-ferrite lamellae), Hu et al. [15] realized the synergistic improvement of mechanical properties, with an UTS (ultimate tensile strength) of 2112 MPa and TE of 16 %, via hot rolling and tempering. Thirdly, the introduction of second-phase particles is widely applied to enhance the mechanical properties of materials, Prasad et al. [16] introduced ultra-fine spherical cementite into the material through multi-pass warm rolling and annealing, achieving a UTS ≥825 MPa and TE of 30 %.

Based on the aforementioned studies, as a core process parameter, rolling temperature directly dominates phase composition, interface characteristics, and defect distribution. Furthermore, existing research has a gap, which concerns the micro-mechanism of spherical carbides in lamellar structures interacting with crack paths to influence toughness. To address this gap, this study aims to investigate the effect of spherical carbides on mechanical properties and impact fracture behavior in lamellar microstructures. Focusing on low-density δ-ferrite steel, this work systematically establishes the relationship among rolling temperature, carbide evolution, and crack propagation behavior through multi-scale microstructure characterization and mechanical property testing. Recently, we have developed DRX kinetics studies through thermo-mechanical deformation experiment [17], where different rolling temperatures were selected to cover the temperature ranges of carbide precipitation/dissolution, DRX, and DIFT. Previous efforts to improve the mechanical properties of δ-ferrite lightweight steel have typically focused on grain refinement of δ-ferrite via DRX. While high-temperature rolling can induce DRX for grain refinement, it also triggers phenomena such as DIFT, exerting unpredictable impacts on material properties. Thus, compared with martensite/δ-ferrite lamellar steel formed by high-temperature rolling, a rolling process that minimally induces DRX and DIFT is urgently required.

The innovation of this study lies in recognizing the phenomenon that second-phase particles dispersedly distributed in the microstructure can enhance the strain hardening capacity of steel, and combining spherical carbide strengthening with lamellar structure strengthening [18], and this novel lamellar structure also meets the requirement for formation at lower rolling temperatures. Furthermore, this integration establishes the correlation between work hardening behavior and microstructure, which links precipitate characteristics to fracture behavior via theoretical models (e.g., Orowan model). The synergistic mechanism of dislocation storage and second-phase strengthening is deduced in reverse by analyzing the features of θ-curves (work hardening rate vs. true strain). The multi-stage toughening mechanism of carbides in the low-temperature rolling zone is revealed through microstructure analysis as follows: ultra-fine spherical carbides induce crack tip blunting, crack deflection, and microvoid nucleation to trigger horizontal crack propagation. These effects dissipate energy and enable the impact energy to exceed 200 J. Meanwhile, the causes of brittle fracture in samples from the high-temperature rolling zone were analyzed. It was found, by combining EBSD (electron backscatter diffraction) analysis with nanoindentation tests, that equiaxed grains of DIF within honeycomb-like martensite lamellae act as “prefabricated crack paths”. Stress concentration also exists at the phase interface, and the two jointly induce brittle fracture.

2. Experimental materials and methods

2.1. Material and processing routes

In this study, the tested steel was fabricated via VIM (vacuum induction melting). Table 1 presents the chemical compositions of major elements in the tested steel. The steel ingot was homogenized at 1200 °C for 4 h, followed by hot forging to a thickness of 40 mm and air cooling to room temperature. As shown in Fig. 1a, specimens were placed in a box furnace, heated to 650 °C, 750 °C, 850 °C, and 950 °C respectively, and held for 2 h. Subsequent hot rolling deformation was conducted with a 60 % reduction ratio, reducing the plate thickness from 40 mm to 16 mm (Fig. 1a). After air cooling, four groups of samples were obtained and named R650, R750, R850, and R950 in sequence. Thermo-Calc software was used to calculate the equilibrium phase diagram and determine the phase composition at each temperature. The experimental design covered the critical temperature ranges for carbide precipitation/dissolution (≤750 °C) as well as DRX and DIFT (≥850 °C). This parameter setting was derived from previous DRX kinetics studies, ensuring that a 60 % deformation ratio could fully achieve microstructure reconstruction.

2.2. Mechanical tests and microstructure study

Mechanical property tests included tensile tests and Charpy impact tests. Tensile tests were performed using rod-shaped specimens with a gauge length of 25 mm and a diameter of 5 mm, which were sampled along the rolling direction (RD) of the plate. The tensile test was conducted at a crosshead speed of 1.5 mm/min. The specimens used for the Charpy impact test were Charpy V-notch samples with dimensions of 10 mm × 10 mm × 55 mm. The notch was machined on the rolled surface, and the impact direction was perpendicular to the RD. The geometry of the Charpy impact specimens and tensile specimens are illustrated in Fig. 1b and c. Multi-scale microstructure characterization was performed using the following techniques. Firstly, for SEM (scanning electron microscopy) and EBSD analyses, the model of the experimental equipment and the sample preparation method were as follows: An Apreo S HiVac SEM equipped with an Oxford Aztec EBSD detector was used, and all observations were conducted on the RD-ND (Normal Direction) plane. After grinding and polishing, SEM samples were etched with a 5 vol % HNO_3-ethanol solution to reveal their microstructural features; EBSD samples were electro-polished using a 5 vol% $HClO_4$-ethanol solution to ensure high-quality diffraction patterns. Secondly, nanoindentation testing was performed using a G200X nanoindenter with a peak load of 5 mN and a loading/unloading rate of 500 μN/s. Calibration of the indenter was conducted prior to testing, and hardness values were determined following the Oliver-Pharr method. To ensure alignment with the underlying phase and grain structure, indentation distribution was validated through EBSD statistics. Thirdly, high-resolution microstructure observation was carried out via a JEM-F200 (CF-HR) TEM (transmission electron microscope). For sample preparation, specimens were ground to 80 μm thickness, then punched into 3 mm-diameter discs. The final thinning step involved twin-jet electropolishing, utilizing a 5 vol% $HClO_4$ acetic acid solution.

3. Results

3.1. Microstructures

The optical microstructures of the test steel at different rolling

Table 1
Chemical composition of the tested steel (wt.%).

C	Si	Mn	Al	Cr	Mo	Ni	Fe
0.35	0.26	1.47	4.50	0.26	0.28	1.47	Bal

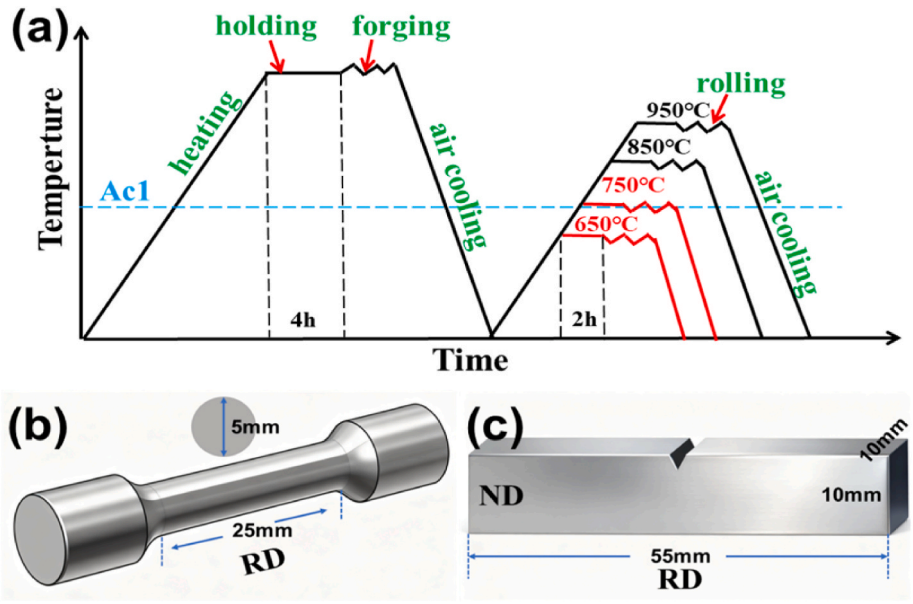


Fig. 1. (a) Schematic representation of the conducted heat treatments for the tested steel, (b) the geometry of tensile specimens and (c) the Charpy impact specimens.

temperatures are shown in Fig. 2. Horizontal and vertical arrows mark RD and ND, respectively. The macrostructure is significantly affected by rolling temperature. In the low-temperature rolling zone, the steel forms a lamellar structure with alternating δ-F (δ-ferrite) lamellae (light areas) and C/F (carbide/ferrite) composite lamellae (dark areas) (Fig. 2a-b). Both types of these lamellae show continuous banded distribution. Table 2 presents the statistics on the proportions of M (martensite), δ-F and C/F lamella in OM (Optical Microscope) images, which were obtained through Image-J software. Data shows that as the rolling temperature increases from 650 °C to 750 °C, the proportion of carbide/ferrite composite lamellae decreases from 51.5 % to 43.3 %, which is consistent with the carbide stability temperature range (<750 °C) predicted by thermodynamic calculations (Fig. 6a). In the high-temperature rolling zone, carbides dissolve completely, and the structure transforms into a duplex structure consisting of δ-ferrite lamellae (light) and honeycomb martensite lamellae (dark) (Fig. 2c-d), with numerous equiaxed ferrite grains (DIF) enclosed within honeycomb martensite lamellae. Notably, as rolling temperature rises from 850 °C to 950 °C, the

Table 2
Phase fraction, carbide size and distribution density of carbides in the tested steel.

Specimen	M [vol %]	δ-F [vol%]	C/F [vol %]	Carbide size [nm]	Distribution density [μm^{-2}]
R650	0	48.5 ± 2.5	51.5 ± 2.5	158.3 ± 76.9	1.7 ± 0.2
R750	0	56.7 ± 3.0	43.3 ± 3.0	247.3 ± 86.8	0.6 ± 0.1
R850	33.2 ± 3.6	66.8 ± 3.6	0	–	–
R950	59.0 ± 4.2	41.0 ± 4.2	0	–	–

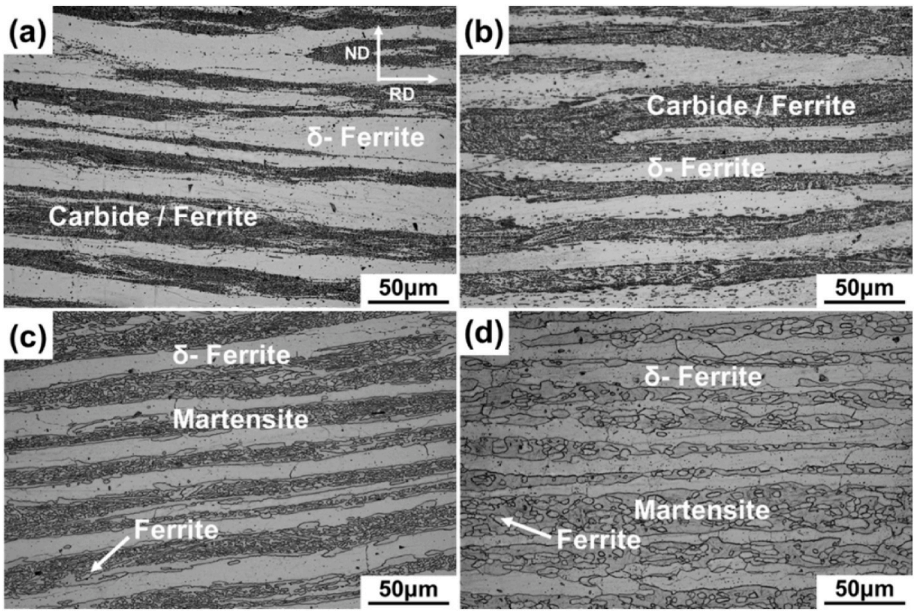


Fig. 2. OM of the tested steel: (a) R650, (b) R750, (c) R850, and (d) R950.

proportion of honeycomb martensite lamellae increases from 33.2 % to 59.0 %, while the equiaxed ferrite grains in martensite grow significantly.

SEM and TEM analyses (Fig. 3) reveal the regulatory mechanism of rolling temperature on the microstructure. As a control group, Fig. 3a-c depict the hot-forged experimental steel prior to rolling. In the figures, there is no obvious lamellar structure in the material before rolling, and the microstructure is mainly composed of coarse δ-ferrite and M/A (martensite/austenite) constituents. In the low-temperature rolling zone (Fig. 3d-i), the characteristics and distribution of carbides can be clearly observed in Fig. 3e. The size distribution histogram of carbides are shown in Fig. 3f and i. Carbides are dispersed as ultra-fine spherical particles in the ferrite matrix of the lamella where they are located, forming ultra-fine spherical carbide/ferrite composite lamellae. By contrast, there are few carbides in the δ-ferrite lamellae, and obvious boundaries exist between the lamellae. In R650 sample (Fig. 3d-f), spherical carbides show a high-density and ultra-fine (average diameter: 158.3 ± 76.9 nm, distribution density: 1.7 ± 0.2 μm^{-2}) within the lamella. When the temperature rised to 750 °C (Fig. 3g-i), the carbides undergo significant coarsening (average diameter increases to 247.3 ± 86.8 nm, distribution density: 0.6 ± 0.1 μm^{-2}) and exhibit a distribution characteristic of "coarse carbides at grain boundaries + fine carbides inside grains", which is consistent with Ostwald ripening [19,20]. In the high-temperature rolling zone (Fig. 3j-o), spherical carbides disappear completely, and the original lamellae are replaced by honeycomb martensite lamellae. It can also be observed that the deformation-induced equiaxed ferrite grains in the honeycomb structure grow with the increase of rolling temperature.

As shown in Fig. 4, Electron Back-Scatter Diffraction and nano-indentation techniques reveal the relationship between microstructure

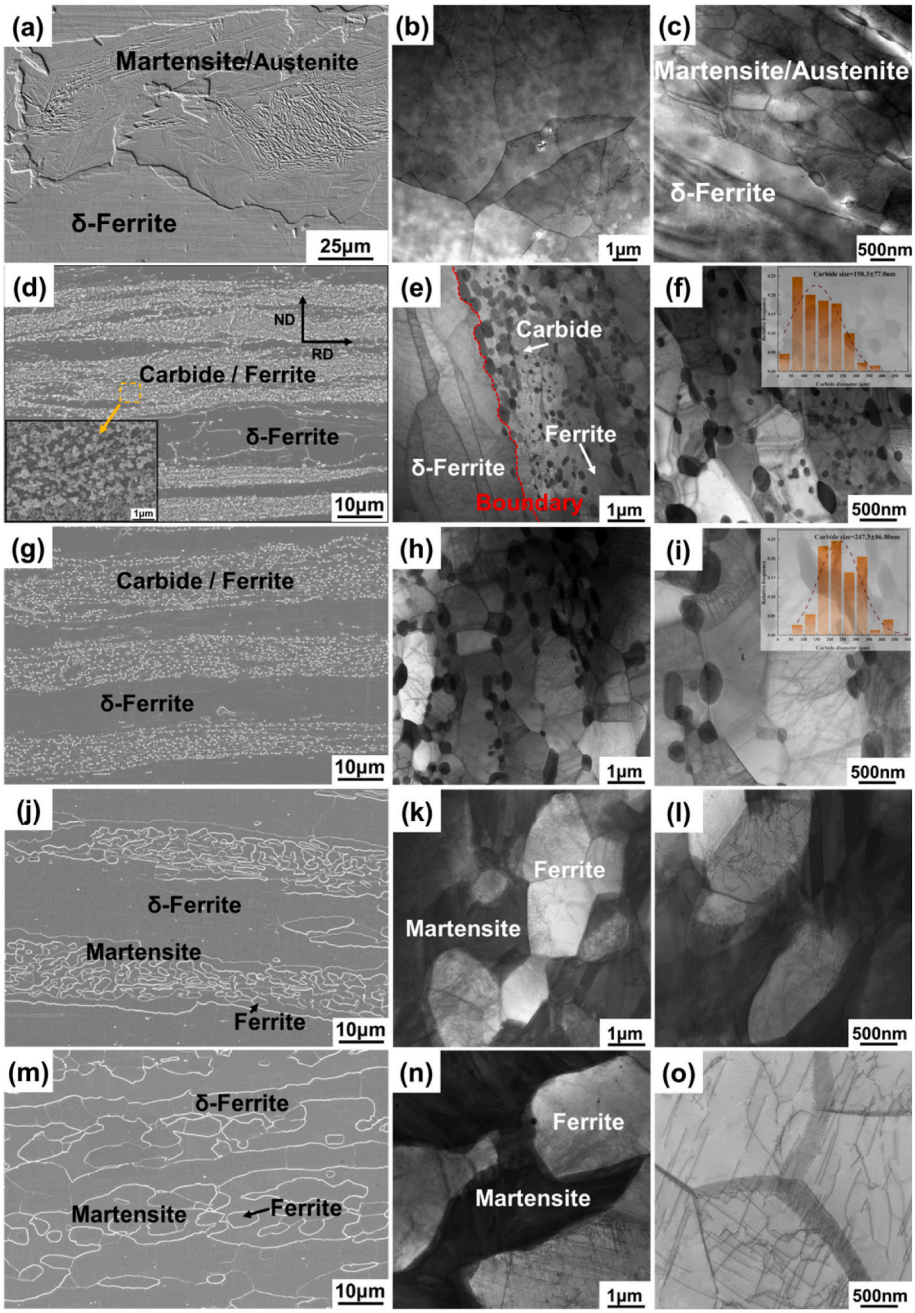


Fig. 3. (a) SEM, (b,c) TEM of the microstructure before hot rolling; (d) SEM, (e,f) TEM of R650; (g) SEM, (h,i) TEM of R750; (j) SEM, (k,l) TEM of R850; (m) SEM, (n, o) TEM of R950.

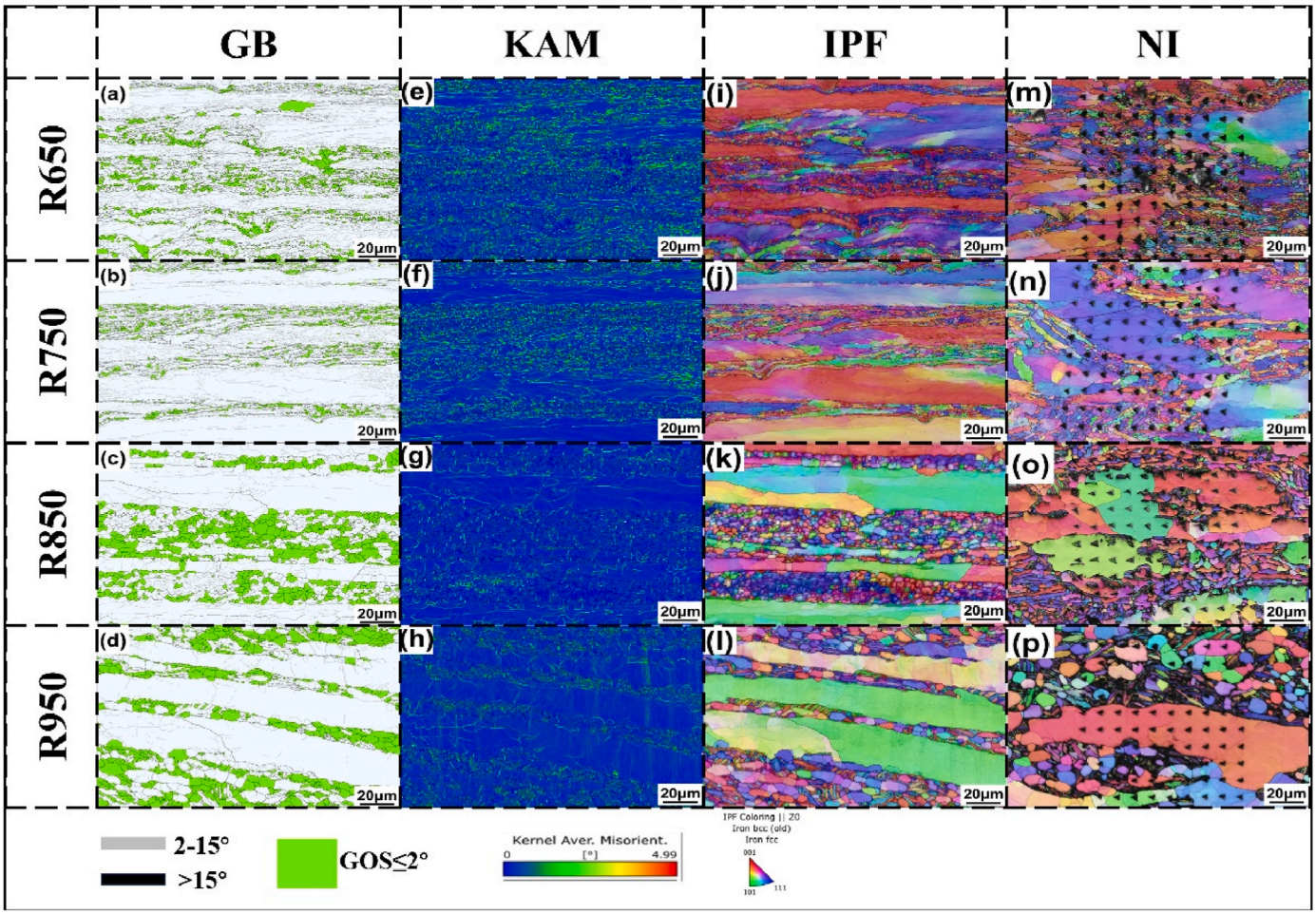


Fig. 4. EBSD maps of the tested steel: (a–d) GB + GOS maps, (e–h) KAM maps, (i–l) IPF (Inverse Pole Figure) maps, and (m–p) IPF maps of nanoindentation.

and mechanical properties. GB (grain boundary) maps are presented in Fig. 4a-d. The proportion of HAGBs (high-angle grain boundaries) in samples from the low-temperature rolling zone ranges from 43.5 % to 49.6 %, while that in samples from the high-temperature rolling zone increases to 60.1 %–75.7 % (Table 3). Samples in the low-temperature rolling zone exhibit an average KAM (kernel average misorientation) value of 0.48–0.55° (Fig. 4e-f), indicating a high dislocation density. In contrast, the KAM value of samples in the high-temperature rolling zone decreases to 0.3–0.4° (Fig. 4g-h). To account for the effects of step size and grain size, the formula [21].

$$\rho = 2\theta/(\mu b)$$

will be used to calculate the dislocation density, where θ is the average KAM value, μ is the step size, and b is the Burgers vector. Comparison of the proportion of GOS (grain orientation spread) < 2° in the material shows that this proportion increases significantly in the high-temperature rolling zone (19.4–25.8 %) compared with the low-temperature rolling zone (6.9–12.3 %). Meanwhile, as the rolling temperature increased, the grain orientation changes (Fig. 4i-l), with the dominant orientation shifting from <001> (predominant in the low-temperature rolling zone) to <101>. This indicates that high-temperature rolling induced the formation of numerous new grains, including recrystallized grains [22] and products of deformation-induced ferrite transformation.

The nano-hardness distribution is shown in Fig. 4m-p. In the low-temperature rolling zone, the hardness of the carbide/ferrite composite lamellae (3.64 GPa) is significantly higher than that of the δ-ferrite lamellae (2.92–2.93 GPa), with a hardness difference of 0.72 GPa. In the high-temperature rolling zone, the hardness of each phase exhibits a gradient distribution, following the order: martensite (5.98–6.50 GPa) > DIF (2.84–2.98 GPa) > δ-ferrite (2.64–2.77 GPa), and the maximum hardness difference reaches 3.86 GPa. This hardness mismatch induces two key effects. In the low-temperature rolling zone, carbide particles pin dislocations via the Orowan mechanism, leading to an increase in the work hardening rate of the composite lamellae. In the high-temperature rolling zone, the large hardness difference between phases causes stress concentration at phase boundaries, which become preferential sites for crack initiation [23].

Table 3
Proportion of HAGBs, KAM values, grain sizes, and nano-hardness of the tested steel.

Specimen	HAGBs [%]	GOS≤2° [%]	Dislocation density [m^{-2}]	C/F [GPa]	M [GPa]	F [GPa]	DIF [GPa]
R650	49.6	12.3	3.9×10^{14}	3.6 ± 0.6	–	2.9 ± 0.3	–
R750	43.5	6.9	3.4×10^{14}	3.6 ± 0.5	–	2.9 ± 0.2	–
R850	60.1	19.4	2.1×10^{14}	–	6.0 ± 0.5	2.6 ± 0.3	3.0 ± 0.3
R950	75.7	25.8	2.8×10^{14}	–	6.5 ± 0.5	2.8 ± 0.2	2.8 ± 0.4

3.2. Mechanical properties

The mechanical properties of the test steel exhibit a significant gradient change under different rolling temperatures (Fig. 5a-d). The low-temperature rolling zone shows excellent strength-plasticity synergy: the room-temperature IAE (impact absorption energy) of the R650 specimen reaches 207.5 ± 29.3 J (Table 4), with an elongation of 19.0 ± 3.2 % and a SPP (strength-plasticity product) as high as 16.51 GPa·%. In the Charpy impact test, the impact load-displacement curve of R650 (Fig. 5b) shows an obvious toughness propagation stage, where the crack Ei (initiation energy) and Ep (propagation energy) reach 59.6 J and 147.9 J, respectively. The crack initiation energy is related to plastic deformation and reflects the difficulty of crack formation, while the crack propagation energy is related to the material's ability to inhibit unstable crack propagation—this indicates that both crack formation and propagation require high energy consumption [24,25]. The impact energy of the R750 specimen decreases to 75.9 ± 22.3 J, but the elongation remains at 17.3 ± 3.6 %. At this point, Ep is greatly reduced to 51.51 J, and the toughness propagation stage is significantly shortened.

The properties of the high-temperature rolling zone deteriorate sharply: the impact energies of R850 and R950 are only 5.6 ± 0.6 J and 4.6 ± 0.2 J, respectively, and their elongations drop to 8.3 ± 0.1 % and 3.5 ± 0.8 % (Fig. 5d). Their impact curves (Fig. 5c) show a steep single peak, with Ei and Ep almost disappearing—this confirms that cracks propagate unstably immediately after initiation. The evolution of tensile properties also confirms this rule. Tensile strength rises steadily as rolling temperature increases from 868.7 MPa in R650 to 1254.5 MPa in R950. However, this comes at the cost of significant losses in plasticity and toughness, and the SPP drops sharply from 16.51 GPa·% to 4.4 GPa·%.

The work hardening rate (θ) curves (Fig. 5e-f) reveal the intrinsic mechanism of property differences. The work hardening curve of the low-temperature rolling zone specimen shows a typical hump shape. In the initial stage ($\varepsilon < 0.02$), it decreases rapidly due to the dominance of dislocation slip; then it rises slightly (strengthening by dislocation interaction), forming a "hump"; finally, it decays slowly when $\varepsilon \geq 0.03$. This phenomenon originates from the synergistic effect of the carbide/ferrite composite lamellae, in which carbides (diameter <250 nm) pin dislocations via the Orowan mechanism, leading to an increase in dislocation density that improves strain hardening capacity and delays necking. In contrast, the specimens in the high-temperature rolling zone exhibit a monotonically decreasing pattern without a rising stage (Fig. 5f), which indicates the lack of coordinated deformation capacity among multiple phases at this temperature.

4. Discussion

4.1. Synergistic effect mechanism of rolling temperature on microstructure

According to the phase diagram (Fig. 6), carbides exist in the material of the low-temperature rolling zone. After high-deformation rolling, the carbides mainly exist in the form of ultra-fine spheres, forming an alternating structure of δ-ferrite lamellae and carbide/ferrite composite lamellae. The formation of this structure mainly originates from the following mechanisms. First is the difference in strain distribution and phase stability. During rolling deformation, the soft-phase ferrite preferentially undergoes plastic flow and elongates significantly along the RD. In contrast, the hard-phase martensite mainly bears elastic deformation and maintains the lamellar outline. This inhomogeneous strain distribution causes the two phases to extend alternately along the thickness direction, forming a lamellar structure parallel to the RD.

As can be seen from Fig. 3a-c, the sample of the tested steel before rolling did not exhibit obvious lamellar structural characteristics before rolling, with the δ ferrite predominating. The large-sized grains are δ-ferrite, and martensite and austenite coexist in the gaps of δ-ferrite, indicating that most austenite transformed into martensite after hot forging. During low-temperature rolling, due to the large deformation, the supersaturated martensite breaks and decomposes under plastic deformation. The deformation energy activates the diffusion of carbon (C), chromium (Cr), and manganese (Mn) atoms, which enrich from martensite to defect sites (dislocations, grain boundaries) and further precipitate carbides [26,27].

As shown in the TEM and energy dispersive spectroscopy (EDS) images of each group of R650 in Fig. 7, significant element enrichment

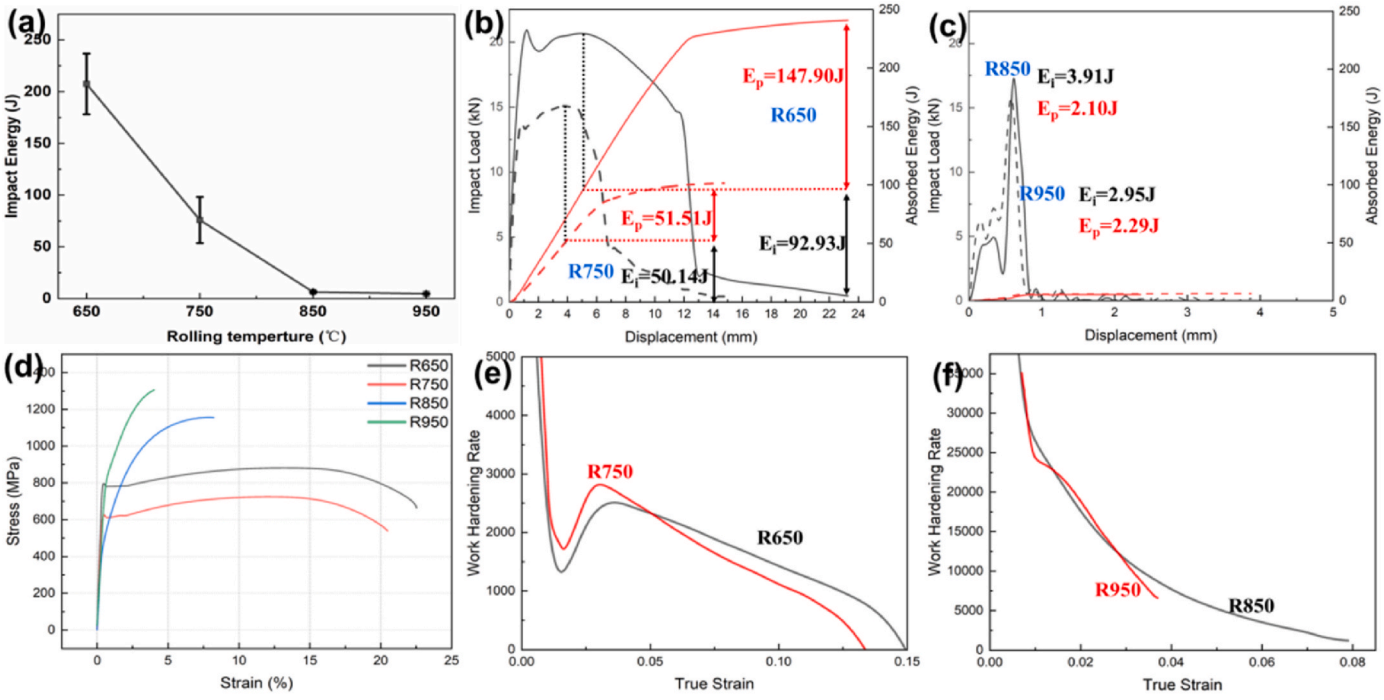


Fig. 5. (a) Room-temperature impact energy absorption, (b, c) impact load-displacement curves, (d) engineering stress-strain curves under room-temperature tension, and (e, f) work-hardening rate curves for the tested steels R650, R750, R850, and R950.

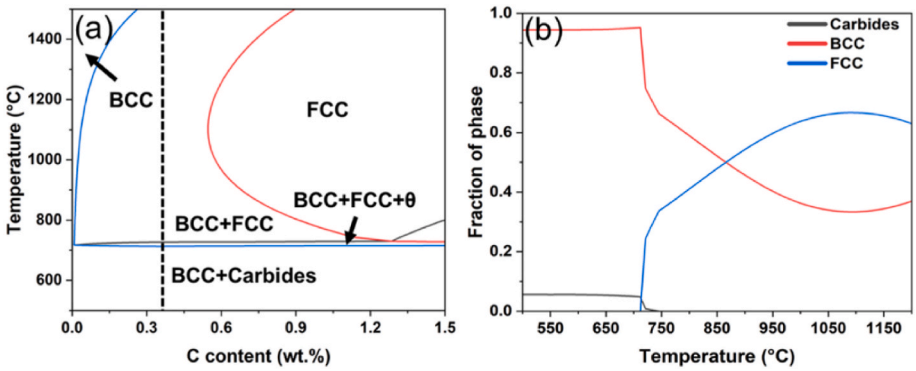


Fig. 6. Phase diagrams: (a) Iron-carbon equilibrium phase diagram; (b) Volume fraction proportion of each phase.

Table 4
Mechanical properties of tested steels.

Specimen	IAE (J)	UTS (Rm, MPa)	TE (A, %)	SPP (Rm × A, GPa %)
R650	207.5 ± 29.3	868.7 ± 16.4	19.0 ± 3.2	16.5 ± 1.5
R750	75.9 ± 22.3	724.3 ± 7.0	17.3 ± 3.6	12.5 ± 2.6
R850	5.6 ± 0.6	1160.0 ± 15.4	8.3 ± 0.1	9.6 ± 0.1
R950	4.6 ± 0.2	1254.5 ± 70.0	3.5 ± 0.8	4.4 ± 1.0

in the carbide regions can be observed. Combined with diffraction pattern calibration, the existence of various carbides such as M_3C, $M_{23}C_6$, and M_7C_3 can be confirmed, and these carbides are mainly concentrated outside the δ-ferrite lamellae. Since δ-ferrite is difficult to undergo phase transformation, it can be concluded that these carbides accumulate in the original martensite regions as products of element enrichment after martensite fragmentation. The massive formation of carbides causes new ferrite to form in the low-carbon regions near them, eventually forming composite lamellae with carbides dispersed in the ferrite matrix.

According to the Ostwald ripening mechanism, an increase in rolling temperature lowers the energy barrier for solute atom migration, thereby facilitating their migration during the ripening process. When the rolling temperature rose from 650 °C to 750 °C, the diffusion coefficients of elements such as C, Mn and Cr increased. Small-sized carbide particles with high interfacial energy dissolve into the ferrite matrix, then diffuse through paths like grain boundaries and dislocations to the surface of large-sized carbide particles with low interfacial energy, where they precipitate to complete the ripening process. The distribution characteristic of "coarse carbides at grain boundaries + fine carbides inside grains" is attributed to the fact that, compared with carbides inside grains, carbides at grain boundaries have higher energy and shorter diffusion paths, which leads to more significant ripening behavior [28]. In the high-temperature rolling zone, high temperature further increases the diffusion coefficients of atoms, significantly accelerating carbide dissolution. This temperature range lies within the δ-ferrite (δ) + austenite (γ) two-phase region, where the stability of austenite is notably enhanced. Meanwhile, austenite-stabilizing elements (such as Mn) expand the austenite phase region and improve its carbon solubility, further promoting carbide dissolution. As a result, no obvious carbides were observed at this rolling temperature, and a microstructure dominated by δ-ferrite lamellae and honeycomb martensite lamellae was finally formed.

In the low-temperature rolling zone, there are numerous deformed

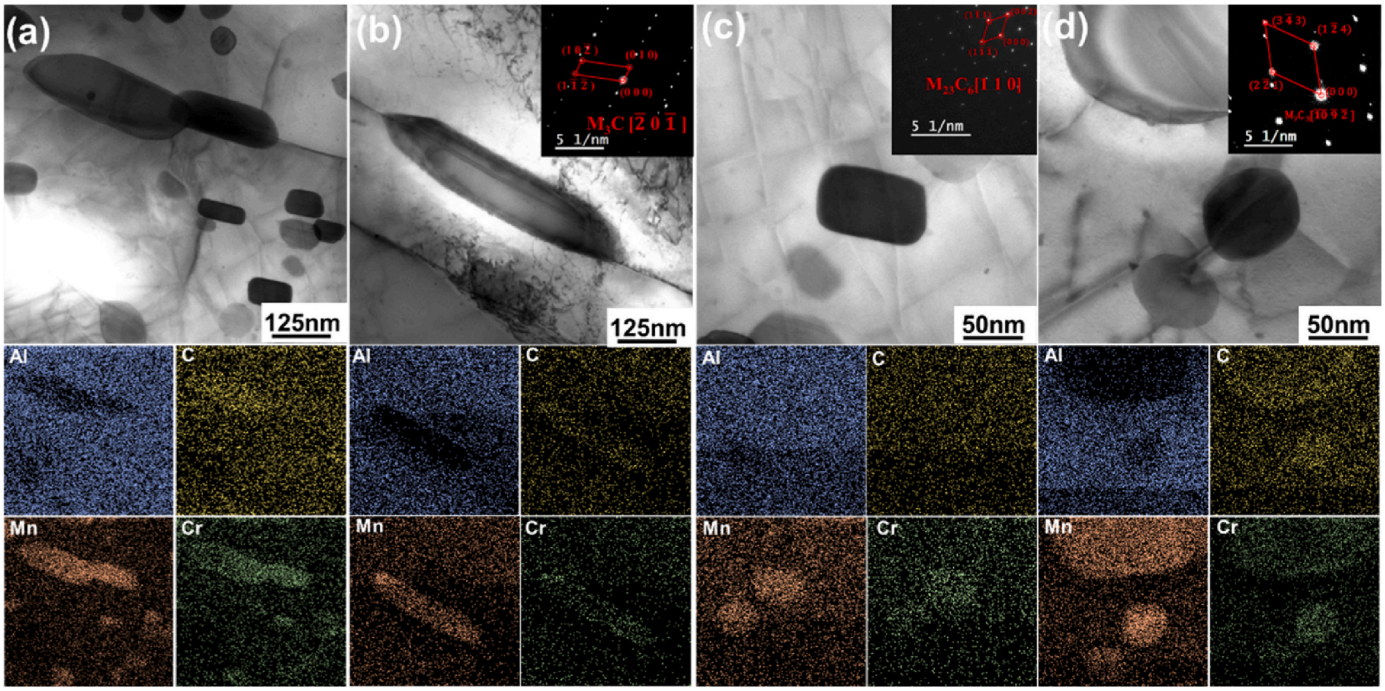


Fig. 7. TEM micrographs and SAED (selected area electron diffraction) patterns, along with EDS results, of the R650 steel: (a) Carbides, (b) M_3C, (c) $M_{23}C_6$, (d) M_7C_3.

grains and a high dislocation density inside the material. The work hardening effect significantly exceeds the dynamic softening effect, leading to a notable increase in the KAM value, which indicates that dynamic recovery (DRV) is the dominant softening mechanism in this temperature zone. In the high-temperature rolling zone, the orientation of numerous grains shifts from <001> to <101>, and new equiaxed ferrite grains are formed. These grains originate from two mechanisms. The first is the dynamic recrystallization process. Combined with EBSD data analysis (Table 3), the proportion of HAGBs and the percentage of grains with GOS $\leq 2^{\circ}$ increased significantly. At the same time, the KAM value and dislocation density decreased. This confirms that subgrain boundaries accumulate orientation differences through dislocation slip and recombination, and finally transform into high-angle grain boundaries, HAGBs thus replaces DRV as the dominant mechanism. And the second is equiaxed ferrite grains that are distributed inside honeycomb martensite lamellae and surrounded by martensite, whose size increases with the rise of rolling temperature. This part of ferrite is attributed to the DIFT occurring during hot rolling. A part of austenite became unstable due to increased plastic deformation storage energy and decomposed into ferrite synchronously during deformation. After rolling, the residual austenite transformed into martensite, resulting in equiaxed ferrite grains being surrounded by the martensite matrix [29,30]. Compared with dynamically recrystallized grains, these DIF have smaller sizes (submicron level) and higher hardness. According to the composition analysis in previous studies [17], the contents of C and Mn in DIF are significantly higher than those in the original δ-ferrite, while the Al content is lower. The solid solution strengthening effect caused by high C content, combined with grain boundary strengthening (Hall-Petch effect), makes its hardness higher than that of recrystallized ferrite.

4.2. *Effect of microstructure on the impact and tensile behaviors*

Experimental results show that the IAE and SPP of the material in the low-temperature rolling zone were significantly higher than that in the high-temperature rolling zone, and there was an essential difference in their fracture mechanisms. The low-temperature rolling zone exhibited ductile fracture, while the high-temperature rolling zone showed brittle fracture characteristics. This performance difference is closely related to the morphology and distribution of the microstructure. Fig. 8 show the OM and SEM images of the deformation zone near the impact and tensile fracture of R650. It can be seen that the crack propagation resistance in the material is mainly borne by the carbide/ferrite lamellae. At this time, the tested steel contained numerous carbides with submicron size (diameter <250 nm). Most carbides were spherical and distributed in the ferrite lamellae, forming a microstructure with alternating arrangement with δ-ferrite lamellae; a small amount existed as short rod-like cementite at the ferrite grain boundaries. In the impact test, cracks initiated and propagated from the V-notch. They exhibited a plane strain state in the δ-ferrite lamellae, as the δ-ferrite grains were coarse and had low hardness—this caused stress concentration at the crack tip and enabled the crack to quickly penetrate this region. When the crack propagated to the carbide/ferrite lamellae, it showed a plane stress state. Interface separation occurred during crack propagation, which dispersed the stress at the crack tip on the interface separation surface, reducing stress concentration and making it difficult for the

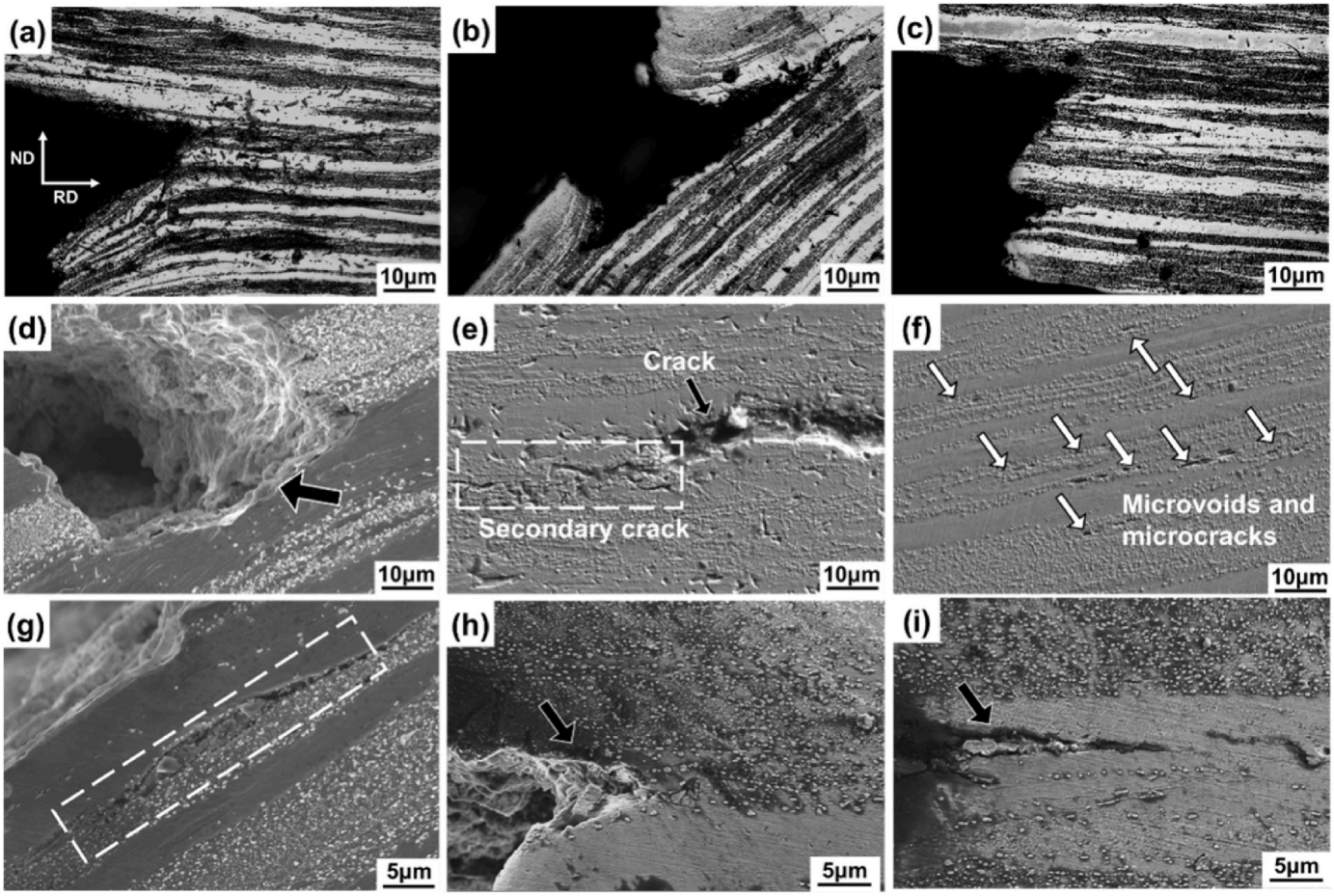


Fig. 8. Microstructural images of the deformed zone in impact fracture surfaces for R650 (a, b) OM, (d–g) SEM; deformed zone in tensile fracture surfaces for R650 (c) OM, (h,i) SEM.

crack to further propagate vertically through the lamellae. Meanwhile, due to high-deformation rolling, the crack needed to pass through numerous lamellae to propagate along the ND, resulting in a significant increase in energy dissipation.

As shown in Fig. 8d, the crack tip became significantly blunted and stopped propagating while penetrating the carbide/ferrite lamellae. This is due to carbides in the lamellae release local stress in the ferrite grains while maintaining good interfacial incompatibility, making it difficult for stress to concentrate in the early stage and for cracks to propagate. According to Griffith's microcrack theory [31] and EET theory [32], the bonding force at the carbide/ferrite interface in the carbide/ferrite lamellae is greater than that at the ferrite/ferrite interface, and the presence of numerous nanoscale spherical carbides can effectively prevent cracks from passing through the carbide/ferrite lamellae. Meanwhile, Du et al. [33] found that precipitated nanoscale spherical carbidess (hard phases) can blunt the crack tip and exhibit excellent crack propagation resistance. Additionally, this phenomenon also originates from the dispersion of stress concentration at the crack tip by spherical carbide particles. When the crack propagates near the spherical particles, the smooth surface of the particles interacts with the crack, increasing the effective curvature radius of the crack tip and thus preventing stress concentration.

According to the theory of elasticity, the stress concentration factor is inversely proportional to the curvature radius—an increase in radius directly reduces the local stress peak at the crack tip, and the synergistic effect of numerous particles enhances the blunting effect [34]. In addition, during the impact process, spherical carbides act as nucleation sites for microvoids (Fig. 8f), inducing the formation of microvoids and microcracks at the carbide/ferrite interfaces. These secondary defects mainly propagate along the horizontal direction of the lamellae (Fig. 8g), releasing the concentrated high stress and thus improving the plasticity and toughness. Compared with the research by Zhang et al. [35], dispersedly distributed spherical carbides can both improve the mechanical properties of materials via the Orowan mechanism. However, unlike the spherical carbides formed by the spheroidization of lamellar cementite in Zhang's study, the nanoscale spherical carbides in this work are derived from martensite decomposition, featuring smaller size, more complete sphericity, and more uniform distribution. Consequently, they better demonstrate the Orowan mechanism.

Microstructural analysis of the tensile deformation zone of low-temperature rolling zone specimens revealed that carbide/ferrite lamellae exerted a significant positive regulatory effect on the material's UTS, TE and SPP. Firstly, the dispersed distribution of ultrafine spherical carbides induced a dispersion strengthening effect, effectively enhancing material strength. Further observation indicated that crack propagation was confined within δ-ferrite lamellae and ran parallel to the lamellar orientation (Fig. 8h-i). The presence of nanoscale carbides enabled carbide/ferrite lamellae to form a barrier-like obstruction [33, 34], greatly improving crack penetration resistance and forcing cracks to propagate along longer paths within δ-ferrite lamellae. On this basis, the ultrafine spherical carbides in the R650 specimen, with smaller size and higher distribution density, exhibited more pronounced dispersion strengthening and crack inhibition effects, making the UTS, TE and SPP of R650 superior to those of R750. Meanwhile, this lamellar structure consisting of hard phases (carbide/ferrite) and soft phases (δ-ferrite) possessed excellent deformation coordination capability [9],namely that the soft δ-ferrite lamellae provided sufficient space for plastic deformation, effectively coordinating the deformation constraints of hard phase lamellae and avoiding local stress concentration. Under the combined action of these mechanisms, cracks extended to varying degrees along δ-ferrite lamellae, substantially increasing energy consumption during crack propagation and thus delaying the fracture process. Eventually, the tensile fractures of the specimens displayed a typical serrated morphology with delamination characteristics (Fig. 8c), which accounted for the excellent comprehensive TE and SPP properties of specimens from the low-temperature rolling zone.

Fig. 9a-d shows the TEM images of the impact-fractured deformation zone of R650, along with the HRTEM (high-resolution transmission electron microscopy) and IFFT (inverse fast fourier transform) images of selected carbide regions. As observed in Fig. 9a and c, after impact deformation, numerous dislocations were blocked by carbides and piled up in the regions adjacent to carbides. Fig. 9b and d are HRTEM images of the carbide regions, where dark areas represent carbides and light areas correspond to the ferrite matrix. Inverse Fourier transform was performed on the ferrite/carbide boundaries to obtain IFFT filtered images. Analysis of the dislocation characteristics reveals that when carbide diameter exceeds 10 nm, numerous dislocations and atomic shearing are observed at the ferrite/carbide boundaries. This is because the Orowan mechanism [36,37] hinders dislocation movement, thereby strengthening the lamellae. Meanwhile, part of the interfacial stress between the ferrite matrix and carbides is transferred to the carbides, alleviating interfacial stress concentration and making it difficult for cracks to propagate through the lamellae containing carbides—this improves the fracture toughness of the material. Based on the above mechanisms, the fracture mode exhibits a mixed pattern (Fig. 9e and h): on one hand, crack blunting makes it difficult for cracks to propagate perpendicular to the lamellae; on the other hand, lamellar separation and secondary defects induce cracks to propagate along the direction of lamellar separation.

In the R750 specimen, the carbide content decreases significantly and the particle size coarsens, as shown in Fig. 3e-f. Although the R750 and R650 specimens have similar microstructural morphologies and comparable hardness of each phase, there is still a significant difference in their impact absorption energy. It can be inferred that the difference in carbides plays a crucial role in the material's fracture behavior at this point. The decrease in carbide proportion and particle coarsening are likely key reasons for the significant weakening of the toughening effect. This change in carbides leads to a reduction in crack deflection resistance, making it easier for cracks to propagate in a straight line and thus lowering energy absorption. It also affects the blunting effect: coarse particles have a weakened ability to disperse stress at the crack tip, and the reduced tip curvature radius causes an increase in the stress intensity factor. Meanwhile, the decrease in carbide quantity results in lower carbide density and larger spacing, reducing the number of sites that hinder dislocation movement and overall weakening the Orowan mechanism. Finally, the number of microvoid nucleation sites in the horizontal direction decreases; fewer ultra-fine carbides lead to less microcrack initiation, weakening the energy dissipation mechanism of secondary cracks. Therefore, the impact absorption energy of R750 decreased by approximately 60 % compared with R650, and its fracture mode is shown in Fig. 9f and i.

The microstructures of R850 and R950 undergo a fundamental transformation, in which carbides are completely dissolved, forming a dual-phase structure consisting of δ-ferrite lamellae and honeycomb martensite lamellae. Moreover, equiaxed grains obtained via DIFT are embedded within the honeycomb martensite lamellae, creating a honeycomb structure (Fig. 9g). At this point, the impact absorption energy of the test steel decreases significantly (<10 J), which is caused by the following aspects. First is interfacial stress concentration. There are significant hardness differences between the martensite phase, δ-ferrite, and DIF, which weakens the bonding at phase interfaces and leads to more concentrated stress at the crack tip. Second is the “crack propagation shortcut” effect of DIF. The equiaxed ferrite grains act as pre-existing defects, guiding cracks to propagate rapidly in a cleavage mode, which significantly reduces energy consumption. Third is the high brittleness of honeycomb martensite lamellae, which inhibits plastic deformation and causes a sharp decrease in fracture toughness. Its fracture schematic is shown in Fig. 9j, which conforms to the characteristics of typical brittle fracture. This confirms that high-temperature rolling significantly degrades material toughness through microstructure coarsening and the formation of brittle phases.

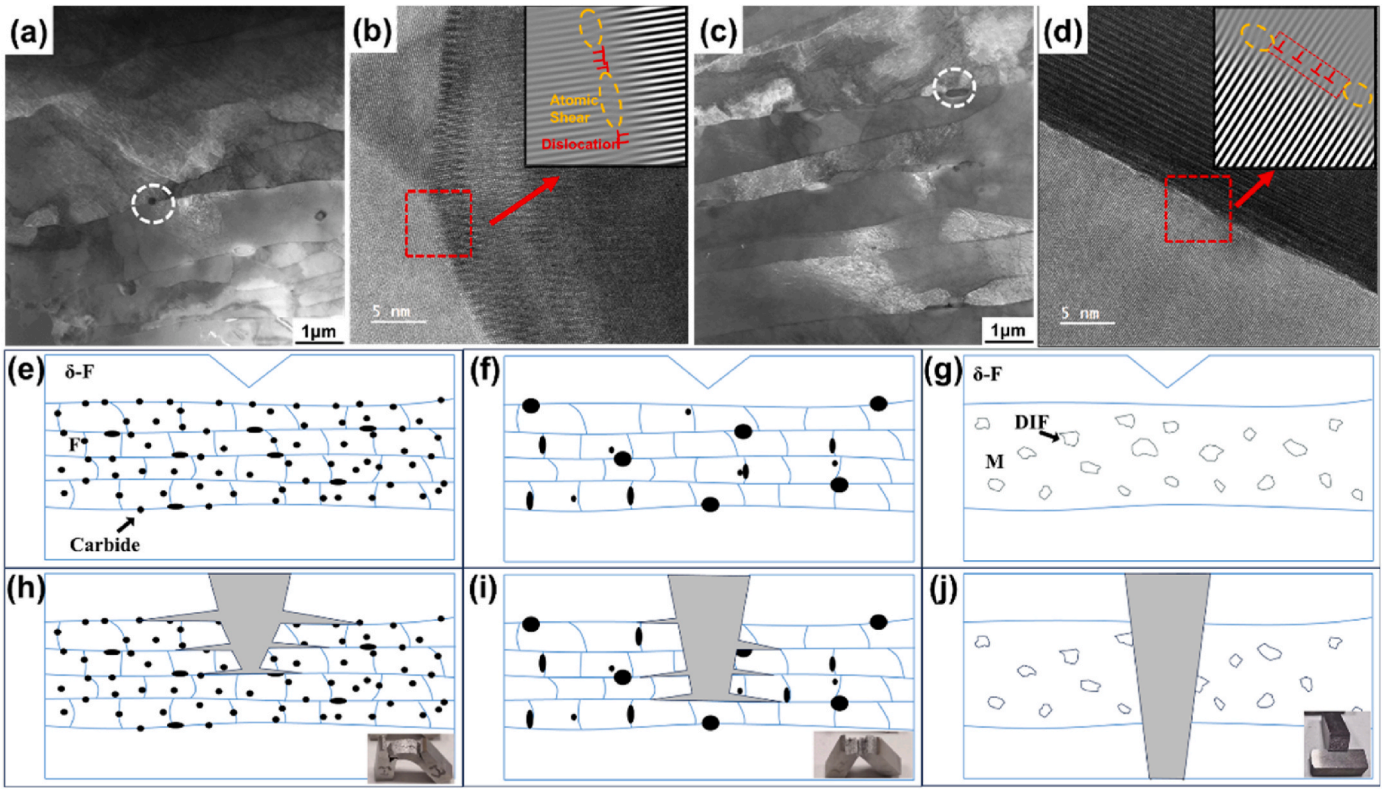


Fig. 9. Images of the deformed zone in impact fracture surfaces for R650 (a,c)TEM, (b, d)HRTEM and IFFT; Schematic diagrams of impact fracture for (e, h) R650, (f, i) R750, and (g, j) R850, R950.

5. Conclusions

1. Through low-temperature rolling, a novel lamellar steel composed of δ-ferrite lamellae and carbide/ferrite lamellae has been prepared. A notable feature of this steel is the presence of numerous nanoscale spherical carbides within the carbide/ferrite lamellae. These carbides inhibit crack propagation during impact and tensile processes, endowing the tested steel with excellent toughness (IAE = 207.5 ± 29.3 J). While reducing the density of the experimental steel, this approach meets the requirement for improved mechanical properties, providing a new strategy for enhancing the mechanical performance of δ-ferrite lightweight steel.

2. The improvement of the tested steel's toughness by the carbide/ferrite lamellae is closely related to the size and distribution density of the spherical carbides. The ultrafine spherical carbides in R650 exhibit a smaller size (158.3 ± 76.9 nm) and a higher distribution density (1.7 ± 0.2 μm^{-2}), resulting in significantly higher toughness compared to those of R750 (size: 247.3 ± 86.8 nm; distribution density: 0.6 ± 0.1 μm^{-2}). A dual-phase structure consisting of δ-ferrite lamellae and honeycomb martensite lamellae was obtained via rolling in the high-temperature rolling region. Numerous equiaxed ferrite grains exist in the honeycomb martensite lamellae, which act as “crack propagation shortcuts” for pre-existing cracks, accelerating crack propagation and significantly deteriorating the toughness of the tested steel.

CRediT authorship contribution statement

Ming Chen: Writing – original draft, Methodology, Investigation, Data curation. **Feng Hu:** Writing – review & editing, Resources, Investigation, Conceptualization. **Songbo Zhou:** Writing – review & editing, Software, Methodology. **Serhii Yershov:** Supervision, Resources, Funding acquisition. **Kaiming Wu:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors are unable or have chosen not to specify which data has been used.

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